

Overview of BLEU score

Subject: BLEU score

1.

When $r \leq c$, the brevity penalty $BP = 1$, meaning that we don't punish long candidates, and only punish short candidates. When $r > c$, the brevity penalty $BP = e^{1-r/c}$

2.

c is the length of the candidate corpus, that is, $c := \sum_{i=1}^M |y^{(i)}|$
 $c := \sum_{i=1}^M |\hat{y}^{(i)}|$ where $|y|$ is the length of y .

3.

Modified n-gram precision Define the modified n-gram precision function to be $p_n(S; S) := \frac{\sum_{i=1}^M \sum_{s \in G_n(y^{(i)})} \min(C(s, y^{(i)}), \max_{y \in S} C(s, y))}{\sum_{i=1}^M \sum_{s \in G_n(y^{(i)})} C(s, y^{(i)})}$
 $p_n(\hat{S}; S) := \frac{\sum_{i=1}^M \sum_{s \in G_n(\hat{y}^{(i)})} \min(C(s, \hat{y}^{(i)}), \max_{y \in S} C(s, y))}{\sum_{i=1}^M \sum_{s \in G_n(\hat{y}^{(i)})} C(s, \hat{y}^{(i)})}$
The modified n-gram, which looks complicated, is merely a straightforward generalization of the prototypical case: one candidate sentence and one reference sentence. In this case, it is $p_n(\hat{y}; y) = \frac{\sum_{s \in G_n(\hat{y})} \min(C(s, \hat{y}), C(s, y))}{\sum_{s \in G_n(\hat{y})} C(s, \hat{y})}$
To work up to this expression, we start with the most obvious n-gram count summation: $\sum_{s \in G_n(y)} C(s, y) = \text{number of n-substrings in } y$
 $C(s, y) = \text{number of n-substrings in } \hat{y} \text{ that appear in } y$

4.

as the word 'the' and the word 'cat' appear once each in the candidate, and the total number of words is two. The modified bigram precision would be $1/1$ as the bigram, "the cat" appears once in the candidate. It has been pointed out that precision is usually twinned with recall to overcome this problem [7], as the unigram recall of this example would be $3/6$ or $2/7$. The problem being that as there are multiple reference translations, a bad translation could easily have an inflated recall, such as a translation which consisted of all the words in each of the references.[8] To produce a score for the whole corpus, the modified precision scores for the segments are combined using the geometric mean multiplied by a brevity penalty to prevent very short candidates from receiving too high a score. Let r be the total length of the reference corpus, and c the total length of the translation corpus. If $c \leq r$, the brevity penalty applies, defined to be $e^{(1-r/c)}$. (In the case of multiple reference sentences, r is taken to be the sum of the lengths of the sentences whose lengths are closest to the lengths of the candidate sentences. However, in the version of the metric used by NIST evaluations prior to 2009, the shortest reference sentence had been used instead.) iBLEU is an interactive version of BLEU that allows a user to visually examine the BLEU scores obtained by the candidate translations. It also

allows comparing two different systems in a visual and interactive manner which is useful for system development.[9]

5.

Final formula There is not a single definition of BLEU, but a whole family of them, parametrized by the weighting vector $w := (w_1, w_2, \dots)$. It is a probability distribution on $\{1, 2, 3, \dots\}$, that is, $\sum_{i=1}^{\infty} w_i = 1$, and $\forall i \in \{1, 2, 3, \dots\}, w_i \in [0, 1]$. With a choice of w , the BLEU score is $BLEU_w(\hat{S}; S) := BP(\hat{S}; S) \cdot \exp \left(\sum_{n=1}^{\infty} w_n \ln \frac{p_n(\hat{S}; S)}{p_n(S)} \right)$. In words, it is a weighted geometric mean of all the modified n-gram precisions, multiplied by the brevity penalty. We use the weighted geometric mean, rather than the weighted arithmetic mean, to strongly favor candidate corpuses that are simultaneously good according to multiple n-gram precisions. The most typical choice, the one recommended in the original paper, is $w_1 = w_4 = 1/4$.

6.

BLEU (bilingual evaluation understudy) is an algorithm for evaluating the quality of text which has been machine-translated from one natural language to another. Quality is considered to be the correspondence between a machine's output and that of a human: "the closer a machine translation is to a professional human translation, the better it is" – this is the central idea behind BLEU.[1] Invented at IBM in 2001, BLEU was one of the first metrics to claim a high correlation with human judgements of quality,[2][3] and remains one of the most popular automated and inexpensive metrics. Scores are calculated for individual translated segments—generally sentences—by comparing them with a set of good quality reference translations. Those scores are then averaged over the whole corpus to reach an estimate of the translation's overall quality. Intelligibility or grammatical correctness are not taken into account.[4] BLEU's output is always a number between 0 and 1. This value indicates how similar the candidate text is to the reference texts, with values closer to 1 representing more similar texts. Few human translations will attain a score of 1, since this would indicate that the candidate is identical to one of the reference translations. For this reason, it is not necessary to attain a score of 1. Because there are more opportunities to match, adding additional reference translations will increase the BLEU score.[5]